

LUNAR ICE INTELLIGENCE & MISSION REPORT

Target: **Faustini Crater (Chandrayaan-2 SAR Swath)** | Data Mode: **REAL** | Chandrayaan-2 Remote Sensing Architecture v2.0

1. EXECUTIVE MISSION SUMMARY

This decision-support analysis establishes landing feasibility, identifies candidate subsurface ice-bearing deposits, and plans hazard-aware rover traverses for **Faustini Crater (Chandrayaan-2 SAR Swath)**. The analysis integrates Chandrayaan-2 DFSAR polarimetric radar (CPR/DOP), LOLA/DEM illumination and terrain hazards, machine learning ice likelihood scoring, and multi-objective A* rover navigation.

Metric	Value	Operational Significance
Candidate Ice Area	0.00 km²	CPR > 1.00 AND DOP < 0.13, over cells carrying radar
Estimated Ice-Equivalent Volume	0 m³	candidate area x 5 m assumed depth x 0.15 assumed pore fraction — an assumption
Recommended Landing Site	Alpha Ridge (North)	Highest composite score (safety + illumination - distance)
Rover Traverse Distance	18.06 km	Science-Aware route, uniform-cost search over slope and hazard
Estimated Rover Energy	3,137.5 Wh	Simplified engineering estimate, 30 kg rover

2. DFSAR RADAR ANALYSIS & ICE INTELLIGENCE

Circular Polarization Ratio (CPR): High CPR (> 1.0) inside permanent shadow regions indicates the Coherent Backscatter Opposition Effect (CBOE), characteristic of low-loss dielectric media such as water ice.

Degree of Polarization (DOP): Depressed DOP (< 0.13) confirms multiple subsurface volume scattering.

Scientific Baseline Screening: Requires CPR > 1.0 AND DOP < 0.13 AND spatial overlap with PSR.

Explainable ML Likelihood: Random Forest model trained on dual-pol and morphological features provides continuous probability distributions without replacing scientific physics.

3. LANDING SITE SELECTION (MULTI-CRITERIA RANKING)

Rank	Site Name	Slope	Hazard	Illumination	Score	Recommendation
#1	Alpha Ridge (North)	6.7°	0.23	0.02	37.1	RECOMMENDED
#2	Beta Plateau (South-East)	8.6°	0.31	0.01	30.0	Alternative
#3	Gamma Bench (South-West)	10.6°	0.37	0.04	27.9	Alternative
#4	Delta Spur (West)	10.7°	0.38	0.03	25.7	Alternative
#5	Epsilon Crest (East)	12.2°	0.43	0.00	22.4	Alternative

4. ROVER TRAVERSE STRATEGY COMPARISON

Strategy	Algorithm	Distance (km)	Mean Hazard	Energy (Wh)	Science Yield
Shortest	Dijkstra	18.06	0.41	3,137.5	0.0
Safest	Dijkstra	34.64	0.24	5,415.7	0.0

Science-Aware	Dijkstra	18.06	0.41	3,137.5	0.0
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5. SCIENTIFIC LIMITATIONS & MISSION ASSUMPTIONS

- 1. Radar Signatures:** *Polarimetric anomalies ($CPR > 1.0$, $DOP < 0.13$) are consistent with potential subsurface ice, but do not constitute certified ground truth confirmation without in-situ drilling or neutron spectrometry.*
- 2. No ML model:** *the Random Forest ice-likelihood classifier was WITHDRAWN. It was fitted to uniformly random labels whose positive class was defined as $CPR\ 1.05\text{-}2.5$ — a range this amplitude-only product cannot reach — so every probability it produced was an extrapolation from fabricated examples. No probability appears in this report.*
- 3. Volumetric uncertainty:** *ice volume is a 3-tier range over assumed regolith stratigraphy (conservative 2 m / 0.05, expected 5 m / 0.15, upper 10 m / 0.3). The area is measured; the depth and the pore fraction are assumptions and are stated with every figure.*
- 4. Rover energy:** *a simplified engineering estimate for a 30 kg micro-rover, not a substitute for certified NASA/ISRO flight dynamic models. The planner is a uniform-cost (Dijkstra) search; no optimality beyond that is claimed.*
- 5. Reproducibility:** *every figure above is computed from the Chandrayaan-2 DFSAR product named on page 1, and re-running the pipeline on that product reproduces them. Any quantity that could not be computed is printed as NO DATA rather than filled in.*